MIL-C-915/58D
18 June 1985
SUPERSEDING
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7 February 1974

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE MWF

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, AWG no. 18, class B stranding, tin coated. Maximum diameter 0.050 inch.

Second - Insulation of synthetic rubber or cross-linked polyethylene. Nominal thickness of 0.020 inch.

Third - Jacket of clear polyamide in accordance with ASTM D 4066, nominal thickness 0.005 inch. Standard identification code applied by method 1.

Fourth - The required number of conductors (see table I) cabled together with a maximum lay of 20 times the pitch diameter of the layer. Cabling sequence to be consecutive, starting with no. 1, from the center outward. Fillers as necessary in order to form a firm, round assembly. Fillers shall be nonfibrous.

Fifth - Binder tape applied helically with overlap.

Sixth - Arctic type neoprene jacket, black. (See table I for thickness.) Cable surface marking required.

FSC 6145

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TABLE I. Details.

Type and size	Number of conductors	Cable jacket thickness nominal (inch)	Overall diameter	
			minimum (inches)	maximum (inches)
MWF-7	7	0.060	0.480	0.500
MWF-10	10	.065	.605	.635
MWF-14	14	.065	.605	.635
MWF-19	19	.085	.710	.745
MWF-24	24	.085	.800	.836
MWF-30	30	.110	.905	.945
MWF-37	37	.110	1.005	1.045

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	7.05
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to water.....	1000
Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor and conductor to water.....	300
Group A:	
Visual and dimensional.....	No failure
Hydrostatic (open end) (on every shipping length) - leakage at 1000 lbf/in^2, in 6 hours, in^3.....	0
Group B:	
Physicals (unaged)	
Insulation (conductor)	
Synthetic rubber	
Tensile strength - lbf/in^2, minimum.....	700
Elongation - percent, minimum.....	350
Cross-linked polyethylene	
Tensile strength - lbf/in^2, minimum.....	1800
Elongation - percent, minimum.....	350
Jacket (cable)	
Tensile strength - lbf/in^2, minimum.....	1800
Elongation - percent, minimum.....	300
Set - inch, maximum.....	3/8
Drip at 95 ± 1°C.....	Zero

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Insulation (conductor)	
Synthetic rubber	
Oxygen pressure	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	75
Air pressure heat	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Air pressure heat	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Cable filler removability	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cable aging and compatibility (125 ± 5°C)	No failure
Cold working (minus 54 ± 2°C)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
7	1000
10	1000
14	500
19	500
24	500
30	500
37	500

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Revision letters are not used to denote changes due to the extensiveness of the changes.

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

Preparing activity:
Navy - SH
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